



5238 - Constraining the Epoch of Reionization via Cross Correlation of JWST Pencil-Beam Galaxy Surveys and Line-Intensity Maps

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Line-intensity mapping is an emerging technique to characterize cosmological large-scale structure through spatial fluctuations in line emission from galaxies and/or the intergalactic medium (e.g., in 21cm or Lyman-alpha). Intensity mapping has the potential to constrain cosmic reionization and the early stages of galaxy evolution by probing numerous faint sources that will be challenging to directly detect, even with JWST. A key challenge in intensity mapping observations is eliminating contamination from astrophysical foregrounds, such as galactic synchrotron emission in 21cm observations, which is orders of magnitude brighter than the cosmological signal. A reliable method of mitigating these contaminants is to cross correlate the intensity mapping signal with a galaxy redshift survey at the same redshift. Recent analytical work has shown that a JWST galaxy survey consisting of a number of pencil beams which sparsely fill the intensity mapping volume can yield a high signal-to-noise when cross correlated with intensity maps such as a Lyman-alpha map from SPHEREx. However, a number of rough assumptions were made in these estimates (e.g., the detection criteria for JWST galaxies and only linear clustering). We propose to address these shortcomings by creating detailed numerical simulations of intensity maps and JWST pencil-beam galaxy surveys within the same volumes. We will survey a wide variety of JWST survey configurations and intensity mapping lines/experiments (e.g., 21cm with SKA and Lyman-alpha with SPHEREx) to determine optimal observing strategies and forecast associated constraints on astrophysical quantities of interest (e.g., timing/topology of reionization).

OBSERVING DESCRIPTION